THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Rule: output $= 3\times$input-2. The output for input 15 is: 1

(a) 44 **(b)** 43 **(c)** 42 **(d)** 45

2. If $a = 6$, $b = 3$, the value of $4a^2 - 8b$ is: 1

(a) 119 **(b)** 120 **(c)** 122 **(d)** 121

3. If $x = 10$, the value of $(x + 6)(x - 3)$ is: 1

(a) 111 **(b)** 114 **(c)** 112 **(d)** 113

4. Perimeter of a rectangle with sides $8x + 2$ and $5x - 1$ is: 1

(a) $6x + 6$ **(b)** $26x + 2$ **(c)** $13x + 1$ **(d)** $80x$

5. If $n = 25$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 327 **(b)** 326 **(c)** 325 **(d)** 324

6. The sum of three consecutive integers starting at n is: 1

(a) $n + 3$ **(b)** $3n + 1$ **(c)** $3n + 3$ **(d)** $3n$

7. If $x = 4$ and $y = 4$, the value of $6x - 3y + xy$ is: 1

(a) 28 **(b)** 27 **(c)** 29 **(d)** 30

8. Rule: output $= 3\times$input-5. The output for input 20 is: 1

(a) 55 **(b)** 54 **(c)** 56 **(d)** 57

9. If $x = 8$, the value of $x^2 - 5x + 6$ is: 1

(a) 30 **(b)** 29 **(c)** 31 **(d)** 32

10. If $x = 9$, the value of $x^2 - 7x + 10$ is: 1

(a) 27 **(b)** 30 **(c)** 29 **(d)** 28

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of $4x^2 - 2x + 6$ at $x = 3$. 2

12. Simplify: $6(x - 4) - 2(x - 7)$. 2

13. Write the sum of the three consecutive multiples of 6, starting with $6n$, and simplify. 2

14. Find the value of $5x^2 - 6x + 7$ at $x = 2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rectangle has length $(8x + 5)$ and breadth $(3x - 2)$ cm. Find the perimeter as an expression and its value for $x = 4$. 3

16. Show that $12n + 9$ is always odd for every whole number n when 12 is even and 9 is odd. Verify for $n = 6$ and $n = 7$. 3

17. Two taxis: A charges ₹40 plus ₹13 per km, B charges ₹100 plus ₹8 per km. Write both fares for k km. For which k are they equal? (Solve $40 + 13k = 100 + 8k$.) 3

OR

A rectangle has length $(3x + 7)$ and breadth $(6x - 4)$ cm. Find the perimeter as an expression and its value for $x = 9$. 3

18. Show that $2n + 11$ is always odd for every whole number n when 2 is even and 11 is odd. Verify for $n = 8$ and $n = 9$. 3

OR

Two taxis: A charges ₹15 plus ₹20 per km, B charges ₹75 plus ₹10 per km. Write both fares for k km. For which k are they equal? (Solve $15 + 20k = 75 + 10k$.) 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In a pattern, figure n has $7n + 4$ tiles. Find 5

(i) tiles in figure 10;
(ii) the figure with 116 tiles;
(iii) the total tiles in figures 1 and 2.

OR

Priya has ₹x. Her brother has ₹18 more than twice her money. Together they have ₹153. Write the expression for the total, form the equation, and find x. Also find the brother's money. 5

20. In a pattern, figure n has $8n + 5$ tiles. Find 5

(i) tiles in figure 9;
(ii) the figure with 109 tiles;
(iii) the total tiles in figures 1 and 2.

OR

Priya has ₹x. Her brother has ₹25 more than twice her money. Together they have ₹205. Write the expression for the total, form the equation, and find x. Also find the brother's money. 5

END OF QUESTION PAPER — Total: 40 Marks

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